Luca Boggiani

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EDUCATION

- **DTU - Università degli Studi di Padova,** Denmark and Italy, *MSc in Electrical Engineering and Bioengineering.* Aug 2023 - Jun 2027
  - **Top International Manager in Engineering,** Italy and Denmark**,** *Double degree student.*

  Study line in Robotics and Autonomous Systems. *GPA 9.8/12.0 (Dan).*
  Study line in Neuroscience and Rehabilitation. *GPA 3.8/4.0 (It).*
  Relevant courses: Computer Vision, Robotics, Linear Control Design, Robust and Fault Tolerant Control , Digital Instrumentation, Machine Learning, Neurorobotics, Signal Processing, Advanced Statistic, Machinery Safety, Robotics and Automation in Pharmaceutical Process, Bioimaging.

- **Università degli Studi di Padova**, Italy, *Bachelor degree in Biomedical Engineering.* Sept 2020 – Jul 2023
  Graduated with 109/110. Thesis: **Kinematic Analysis of a Robotic Arm in the Motion Plane**

WORK EXPERIENCES

- **Student Assistant – Robotics and Automation,** *NIL Technology,* Copenhagen Aug 2025 - Present
  Working on the control and programming of a Universal Robots (UR) robotic arm to perform high-precision tasks in coordination with laser systems, cameras and power meters. Focusing on motion planning and accuracy optimization to support advanced optical experiments.

- **Visiting Research Intern,** *BioRobotics Group, CAR-CSIC,* Madrid Oct 2026 - Dec 2026
  - Porting a bilateral gait-phase-triggered FES system (IMU-based gait detection, 8-channel Hasomed P24) from PyQt6 to ROS2 and integrating it with a lower-limb exoskeleton through a shared gait-state topic for co-adaptive control.
  - Validating end-to-end latency (<15 ms target) and safety mechanisms on the bench, followed by pilot gait trials comparing exoskeleton-only, FES-only, and integrated conditions.

- **Junior researcher,** *Whi Students team,* DEI Università degli Studi di Padova, Italy Jan 2024 - Oct 2024
  Led a multidisciplinary team in the world's top neurorobotics competition.
  - Led a multi-disciplinary team to develop and test a BCI-controlled robotic arm; responsible for planning and analyzing laboratory test data and documenting outcomes.

- **Member of the Electrical team**. *Float Forward DTU***,** Copenhagen, Denmark Sept 2024 - Jan 2025
  Contributed to the 12th Monaco Energy Boat Challenge entry.
  - Assisted in the design and integration of embedded control systems and cooling system; collaborated on component testing and debugging for electric propulsion system.
- **Intern,** *Moveo,* Padova, Italy. April 2023- June 2023
  Construction of a passive orthosis to promote the gait cycle in patients with neuropathies.
  - Conducted biomechanical tests to assess and optimize device performance.

LANGUAGES AND IT SKILLS

Languages: English (fluent), Italian (native), Spanish (fluent).
IT Skills: Microsoft Office, C, C++, Python, MATLAB, ROS, Unity, CAD, Git VersionControl.
GitHub: https://github.com/243135-tech

PROJECTS

BCI development for an exo control at *NRT-LAB DTU Electro.* Developed and prototyped a Motor Imagery Brain-Computer Interface (BCI) leveraging EEG signals and machine learning in Python and MATLAB. Implemented real-time decoding of imagined limb movements to control a 2-DoF robotic arm and designed an interactive neurofeedback game to enhance user focus and training outcomes. Integrated both the BCI and game with an upper-limb exoskeleton to support active rehabilitation. Awarded with the invitation to the Brain & Technology Symposium 2025 at Hellerup Hospital. *Tools:* Python, MATLAB, EEG, Machine Learning, Robotics [GitHub Link]

Object recognition for autonomous vehicle. Designed and implemented a perception system for autonomous navigation in dynamic environments. Developed algorithms for object detection, classification, and occlusion-aware tracking using Kalman filtering. Enabled continuous multi-entity tracking even under visual occlusion by predicting trajectories. *Tools:* Python, OpenCV, Kalman Filter. [GitHub Link]

INTEREST AND ACHIEVEMENTS

3rd Place at BCI Cybathlon 2024 and Jury award for the most innovative technology, Zurich Oct 2024

2nd Place at Robotech competition, Novo Nordisk Engineering Jan 2026

Finalist of the Danish DTU Robocup 2025, Copenaghen April 2025

Semiprofessional Rugby player and current Captain, Italy - Denmark Sept 2011- Present

REFERENCES

Lazaros Nalpantidis (DTU Electro)**,** Silvia Tolu (DTU Electro)**,** Luca Tonin (DEI Unipd)